In classical approaches to data management, metadata is gathered in dedicated tools, most often in a data catalog, where the submitted dataset needs to be enriched with additional information. In this way, metadata is separated from the dataset, which often leads to metadata descriptions that are not fully synchronized with the current form of data.

The data mesh approach assumes that the data product is autonomous and self-described, in a way that the self-serve platform and end users can use it for different purposes. Therefore, it is recommended to use the *metadata as code* approach, which results in data that will be easily interpreted by machines and validated and versioned using existing source code–oriented tools.

For example, a metadata description can be implemented using JSON notation. The data product's metadata description file contains sections dedicated to each of its aspects. An example file is in the following listing.

Listing 5.1 Sample data product metadata file outline

```
{
"dataProduct": {
   // data product details
},

"domainDataSets": [
   // domain dataset details
],

"schemas": [
   // dataset schema details
],

"terminology": {
   // vocabularies
},
"security": {
   // security definition
},
```